

Modeling cardiac muscle fibers in left ventricular electrophysiology simulations

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1 Introduction and Objectives

In ventricular electrophysiology (EP), the assumption of an isotropic domain leads to non-physiological simulations of action potential propagation. Cardiac tissue is inherently anisotropic: the electrical signal travels significantly faster along the longitudinal axis of the cardiomyocytes compared to the transversal directions. Therefore, incorporating a realistic spatial definition of the myocardial fiber and sheetlet arrangement is a fundamental prerequisite for more accurate and predictive EP modeling.

The primary objective of this project is to implement a *Laplace-Dirichlet Rule-Based Method* (LDRBM), following the algorithm formalized by [2] and recalled by [1], to generate a ventricular fiber field, the so-called *Rossi Rule-Based Method* (R-RBM). To evaluate and validate the algorithm incrementally, the method has been implemented first on a simple cubic domain, then extended to an idealized left ventricular geometry, and finally applied to a realistic left ventricle anatomy. Ultimately, we aim to analyze the generated fiber field and assess its impact on electrical activation by comparing isotropic and anisotropic activation maps.

2 Mathematical Model and Assumptions

In this section, we define the mathematical framework underlying the classical Rossi Rule-Based Method (RBMs), which is a computational algorithm designed to assign spatial fiber orientations across the myocardium by applying a set of mathematically defined rules, which are not derived from physiological and histological observations. These methods bypass the need for complex and often unavailable patient-specific imaging data, such as Diffusion Tensor Imaging (DTI). LDRBMs represent a robust class of RBMs that establish a local coordinate system at every point in the myocardium by solving Laplace equations with appropriate Dirichlet boundary conditions.

The objective is to first define the mathematical framework underlying the R-RBM in order to further apply it to single-ventricle geometries. In particular, we focus on the algorithm originally proposed by Rossi et al. [2] and reviewed by Piersanti et al. [1].

2.1 Rule-based-methods for ventricular fibers generation

The ventricular Rossi Rule-Based Method computes the myocardial fiber architecture through a structured sequence of mathematical operations. The workflow can be conceptually divided into three main phases:

1. **Anatomical definition of the domain:** Let $\Omega_{myo} \subset R^3$ be the continuous spatial domain representing the left ventricular myocardium. The boundary of this domain, $\partial\Omega_{myo}$, is partitioned into mutually exclusive regions based on physiological anatomy: the endocardium Γ_{endo} , the epicardium Γ_{epi} , and the base Γ_{base} . Thus, the complete boundary is mathematically defined as $\partial\Omega_{myo} = \Gamma_{epi} \cup \Gamma_{endo} \cup \Gamma_{base}$.

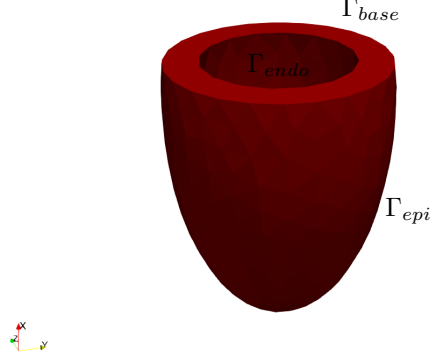


Figure 1: Schematic representation of the left ventricular boundary partition: Γ_{endo} (endocardium), Γ_{epi} (epicardium), and Γ_{base} (base).

2. **Transmural Field:** To establish a local reference system across the myocardium, the method requires computing a normalized transmural distance, denoted as ϕ , by solving a Laplace-Dirichlet boundary value problem:

$$\begin{cases} -\Delta\phi = 0 & \text{in } \Omega_{myo} \\ \phi = 0 & \text{on } \Gamma_{endo} \\ \phi = 1 & \text{on } \Gamma_{epi} \\ \nabla\phi \cdot \mathbf{n} = 0 & \text{on } \Gamma_{base} \end{cases} \quad (1)$$

where $\mathbf{n}$ is the outward normal direction to the base. Once the scalar field ϕ is computed, its gradient directly defines the transmural direction vector $\nu = \nabla\phi$ throughout the ventricular wall.

3. **Local Basis Generation and Physiological Rotation:** An initial orthonormal coordinate system is constructed starting from the definition of the normal direction $\mathbf{k}$, which is given by using the outward normal to the ventricular base, that is $\mathbf{k} = \mathbf{n}_{base}$.

The definition of this orthonormal local coordinate system (P) is given for each point of the domain Ω_{myo} and it's computed using the following function:

$$P = [\hat{\mathbf{e}}_l, \hat{\mathbf{e}}_n, \hat{\mathbf{e}}_t] = \text{axis}(\mathbf{k}, \nu) = \begin{cases} \hat{\mathbf{e}}_t & = \frac{\nu}{\|\nu\|}, \\ \hat{\mathbf{e}}_n & = \frac{\mathbf{k} - (\mathbf{k} \cdot \hat{\mathbf{e}}_t)\hat{\mathbf{e}}_t}{\|\mathbf{k} - (\mathbf{k} \cdot \hat{\mathbf{e}}_t)\hat{\mathbf{e}}_t\|}, \\ \hat{\mathbf{e}}_l & = \hat{\mathbf{e}}_n \times \hat{\mathbf{e}}_t, \end{cases} \quad (2)$$

which takes as input the normal direction $\mathbf{k}$ and the transmural direction vector ν defined before, returning the orthonormal system P composed by the three orthonormal directions $\hat{\mathbf{e}}_l, \hat{\mathbf{e}}_n, \hat{\mathbf{e}}_t$ which, respectively, represent the longitudinal, the normal and the transmural unit directions. In conclusion, for the R-RBM we have that $P = \text{axis}(\mathbf{k}, \nabla\phi)$.

Finally, to replicate physiological features, this local coordinate system must be rotated to match the true fiber and sheet orientations. In order to achieve that, the vectors of the orthonormal system are rotated by suitable angles:

- the longitudinal direction of the myocytes $\hat{\mathbf{e}}_l$ rotates counter-clockwise around $\hat{\mathbf{e}}_t$ by an angle α .
- the transmural direction rotates counter-clockwise around $\hat{\mathbf{e}}_l$ by an angle β .

So, the vectors are rotated to produce a new mapped coordinate system $[f, n, s]$, corresponding to the fiber, cross-fiber, and sheet directions. The rotational angles vary linearly as a function of the normalized

transmural distance ϕ . In particular, the fiber angle $\alpha = \alpha(\phi)$ is a function of the transmural normalized distance $\phi \in [0, 1]$. The fiber angle α is defined as follows:

$$\alpha(\phi) = \alpha_{epi}(1 - \phi) + \alpha_{endo}\phi$$

where α_{endo} and α_{epi} represent the prescribed angles at the endocardium and epicardium. To accurately model the characteristic rotation of the fibers, these boundary angles are assigned opposite signs, in particular in the simulations, in order to match histological observations, the values $\alpha_{endo} = -60^\circ$ and $\alpha_{epi} = +60^\circ$ have been used.

Once the transmural distance ϕ and the helix angle $\alpha(\phi)$ are computed, the continuous cardiac fiber field $\mathbf{f}$ is constructed exploiting the local orthonormal triplet at each point of the myocardium. The local fiber orientation $\mathbf{f}(\phi)$ is obtained by rotating the longitudinal vector $\hat{\mathbf{e}}_l$ around the transmural axis $\hat{\mathbf{e}}_n$ by the angle $\alpha(\phi)$:

$$\mathbf{f}(\phi) = \cos(\alpha(\phi)) \hat{\mathbf{e}}_l + \sin(\alpha(\phi)) \hat{\mathbf{e}}_n \quad (3)$$

This formulation guarantees a smooth vector field.

3 Numerical Results

To evaluate and verify the Rossi Rule-Based Method, the entire framework was implemented in **Python** using the **Firedrake** library. Visualizations and streamline tracing were performed using ParaView and PyVista.

The complete code used to generate these results is available at:

<https://github.com/giuliagambini/Cardiac-Modeling-Fiber-Field-Generation-Project.git>.

The generation of the myocardial fiber architecture followed an incremental pipeline. By successfully testing the method and the rotation first on a canonical geometry then expanding to idealized and complex anatomical structures. Moreover, in all three test cases, the continuous transmural rotation angle $\alpha(\phi)$ was prescribed to vary from an endocardial angle of $\alpha_{endo} = -60^\circ$ to an epicardial angle of $\alpha_{epi} = +60^\circ$, replicating classical histological literature.

3.1 Preliminary Validation on a Cubic Domain

The initial phase of numerical simulation involved a simplified benchmark using a regular continuous three-dimensional unit cube ($\Omega = [0, 1]^3$). In this canonical domain, the bottom face ($\phi = 0$) represents the endocardium, the top face ($\phi = 1$) represents the epicardium, and one of the lateral faces represents the base.

To obtain the solution for ϕ , we translate the continuous Laplace problem into its weak formulation: find $\phi \in H^1(\Omega)$ such that:

$$\int_{\Omega_{myo}} \nabla \phi \cdot \nabla v \, d\Omega = 0 \quad \forall v \in H^1(\Omega) \quad (4)$$

subject to the Dirichlet boundary conditions $\phi = 0$ at Γ_{endo} and $\phi = 1$ at Γ_{epi} . The system is discretized using the Finite Element Method (FEM) within the Firedrake environment. This approach allows us to compute the transmural distance ϕ numerically, yielding a linear scalar field that serves as the basis for the fiber direction vector field.

Due to the non-homogeneous Dirichlet boundary conditions, we define the trial space V_ϕ and the test space V_0 as specific subspaces of the Sobolev space $H^1(\Omega_{myo})$:

$$V_\phi = \{u \in H^1(\Omega_{myo}) : u|_{\Gamma_{endo}} = 0, u|_{\Gamma_{epi}} = 1\} \quad (5)$$

$$V_0 = \{v \in H^1(\Omega_{myo}) : v|_{\Gamma_{endo}} = 0, v|_{\Gamma_{epi}} = 0\} \quad (6)$$

The variational problem reads: find $\phi \in V_\phi$ such that:

$$\int_{\Omega_{myo}} \nabla \phi \cdot \nabla v \, d\Omega = 0 \quad \forall v \in V_0 \quad (7)$$

It is worth noting that a homogeneous Neumann boundary condition $\nabla\phi \cdot \mathbf{n} = 0$ is implicitly enforced on the basal boundary Γ_{base} , representing the structural insulation at the base of the ventricle.

The system is discretized using the Finite Element Method (FEM) with first-order continuous Lagrange elements within the Firedrake computational framework. With this approach the transmural distance ϕ is computed numerically, yielding a continuous scalar field needed for the generation of the rule-based fiber structure.

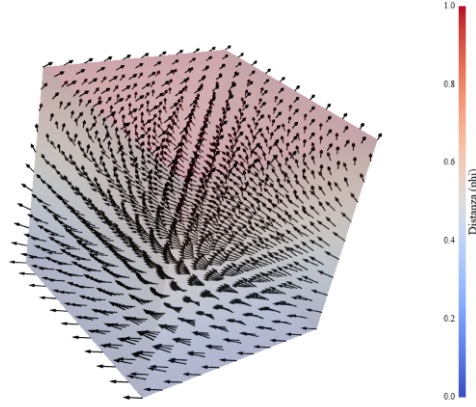


Figure 2: Generated fiber field on the canonical cubic mesh, twist from -60° at the base ($\phi = 0$) to $+60^\circ$ at the top ($\phi = 1$).

The complete orthotropic triad was explicitly computed for the cubic tissue block to demonstrate the full theoretical framework of the R-RBM expanded as in Piersanti et al. [1], but for the idealized and realistic ventricular geometries the algorithm focused exclusively on generating the primary myofiber field $\mathbf{f}$. This streamlined approach is perfectly coherent, as $\mathbf{f}$ is the sole structural parameter required to construct the isotropic conductivity tensor for our electrophysiological simulations.

3.2 Application to an Idealized Ventricular Mesh

Following the cubic validation, the R-RBM framework was applied to a truncated ellipsoid in order to represent an idealized ventricle. This setup serves as a more demanding testbed, characterized by non-trivial surface curvatures and a spatially varying gradient field.

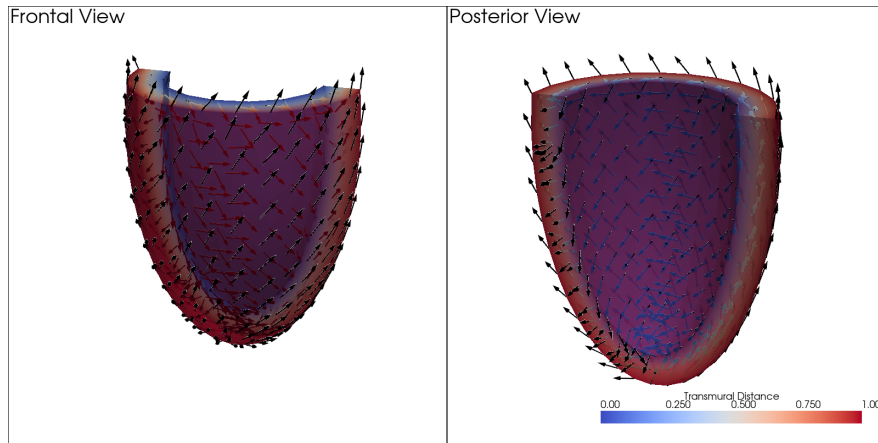


Figure 3: Fiber field on the idealized truncated ellipsoid model, with transmural distance.

The computed scalar potential ϕ naturally follows the ellipsoid, defining a normalized radial thickness coordinate ranging from 0.0 to 1.0 (Figure 3, left). By combining the transmural direction $\hat{e}_t$ with the longitudinal axis $\mathbf{k}$, we built a local reference coordinate frame. Finally, the algorithm successfully generated a dense network of fibers wrapping smoothly around the curved walls (Figure 3, right), confirming its ability to handle also 3D geometries.

3.3 Realistic Anatomical Ventricle

The final challenge for the rule-based method was the irregular left ventricular mesh containing significant local variations in wall thickness, asymmetries and complex boundary regions.

The effectiveness of the Laplace-Dirichlet formulation lies in its PDE-based nature, because the scalar field ϕ is the solution to a diffusion-like Laplace equation, it inherently adapts to the local geometry. This allows the local coordinate frame to stretch and compress automatically according to the myocardium's thickness through the value of ϕ .

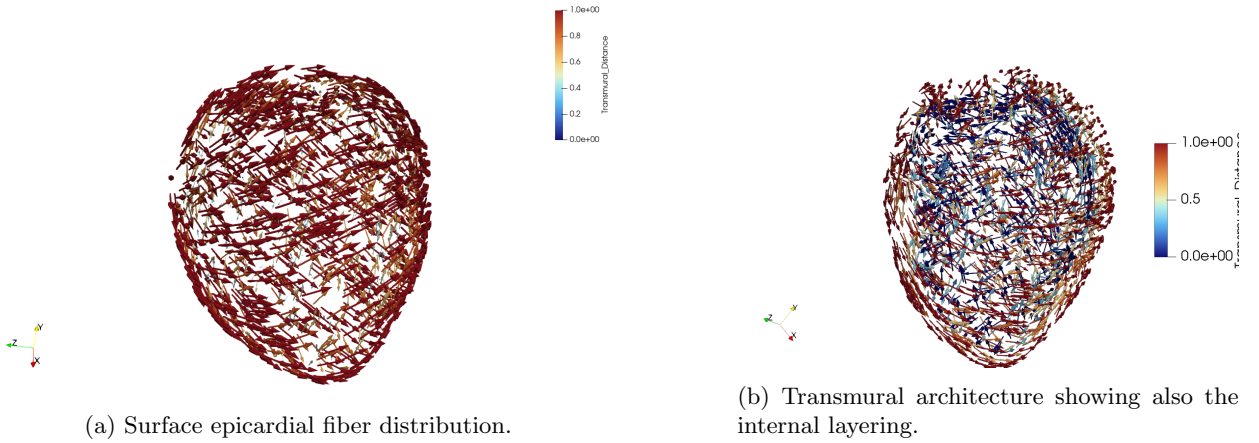


Figure 4: Streamline render ($n = 3000$ density) of the generated fiber field on the left ventricular mesh.

To visualize the resulting fiber architecture, a dense network of streamlines ($n = 3000$) was computed. As shown in Figure 4, the R-RBM successfully generates a continuous, helical structure. Furthermore, a vortex pattern is visible in figure 5.

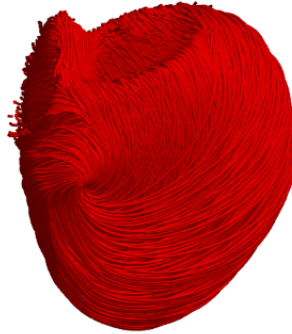


Figure 5: View of the left ventricle showing the smooth and continuous helical curling of the generated muscle fibers, converging at the minimum.

4 Modeling cardiac electrophysiology

To evaluate the impact of the generated fiber field on the electrical propagation, the action potential has been simulated using the monodomain equation which, given the domain Ω_{myo} and a time interval $(0, T]$, reads:

find, for each t , the transmembrane action potential $V : \Omega_{myo} \times (0, T] \rightarrow \mathbb{R}$ such that

$$\chi C_m \frac{\partial V}{\partial t} = \nabla \cdot (\mathbf{D} \nabla V) - \chi I_{ion}(V, \mathbf{w}) + I_{app}$$

where χ is the membrane surface-to-volume ratio, C_m the membrane capacitance per unit area, I_{ion} represents the non-linear transmembrane ionic currents governed by cellular state variables $\mathbf{w}$, and I_{app} is an externally applied stimulating current.

4.1 Model Simplification and Assumption

In this work, the primary objective is to implement and validate the R-RBM for fiber architecture rather than to capture the cellular membrane dynamics. Therefore, to minimize the computational effort, we introduce a key simplification: we neglect the cellular response by setting the ionic current term $I_{ion} = 0$. Furthermore, instead of applying a continuous external current source I_{app} , we initiate the propagation via a localized initial condition. By scaling out physical constants for simplicity, the system reduces to a parabolic diffusion equation:

$$\frac{\partial V}{\partial t} = \nabla \cdot (\mathbf{D} \nabla V) \quad \text{in } \Omega_{myo} \times (0, T]$$

subject to homogeneous Neumann boundary conditions on the entire boundary, representing an electrically insulated domain:

$$(\mathbf{D} \nabla V) \cdot \mathbf{n} = 0 \quad \text{on } \partial\Omega_{myo}$$

The potential $V(\mathbf{x}, t)$ is initialized at $t = 0$ with a spherical stimulus centered at the ventricular minimum point $\mathbf{x}_0$:

$$V(\mathbf{x}, 0) = \begin{cases} 1.0 & \text{if } \|\mathbf{x} - \mathbf{x}_0\| \leq R \\ 0.0 & \text{otherwise} \end{cases}$$

where R is the radius of the activation sphere (set to 5.0 in our implementation).

4.2 Anisotropic and Isotropic Comparison

The activating influence of the myocardial fibers is entirely incorporated into the conductivity tensor $\mathbf{D}$. We assume that the conduction properties are isotropic in the plane perpendicular to the fiber direction for simplicity, so the anisotropic conductivity tensor $\mathbf{D}_{ani}$ is constructed using the local unit fiber vector $\mathbf{f}$ generated by the R-RBM and assuming transversal anisotropy ($\sigma_t = \sigma_n$):

$$\mathbf{D}_{ani} = \sigma_t \mathbf{I} + (\sigma_l - \sigma_t)(\mathbf{f} \otimes \mathbf{f}).$$

where σ_l is the conductivity along the fiber direction and σ_t is the conductivity transverse to the fibers.

In our simulations, we select $\sigma_l = 60.0$ and $\sigma_t = 6.0$, reflecting a $\sigma_l/\sigma_t \approx 10$ ratio, it follows that the velocity ratio is $\sqrt{10} \approx 3.16$. To assess the impact of the fiber architecture, we compare this physiological scenario against an isotropic conduction model, where the tensor is defined as:

$$\mathbf{D}_{iso} = \sigma_t \mathbf{I}$$

This isotropic model completely ignores the fiber orientation, allowing the signal to diffuse uniformly in all directions, similarly as a spherical expansion.

Finally, the diffusion equation is discretized in space using the Finite Element Method with first-order continuous Galerkin elements on the ventricular mesh, and in time using a first-order implicit Backward Euler

scheme with $\Delta t = 0.5$. The resulting system is solved at each step in Firedrake using a Conjugate Gradient solver.

Activation results: Isotropic vs. Anisotropic Propagation

To assess the functional impact of the fibers, we compared the activation time maps under isotropic and anisotropic conditions.

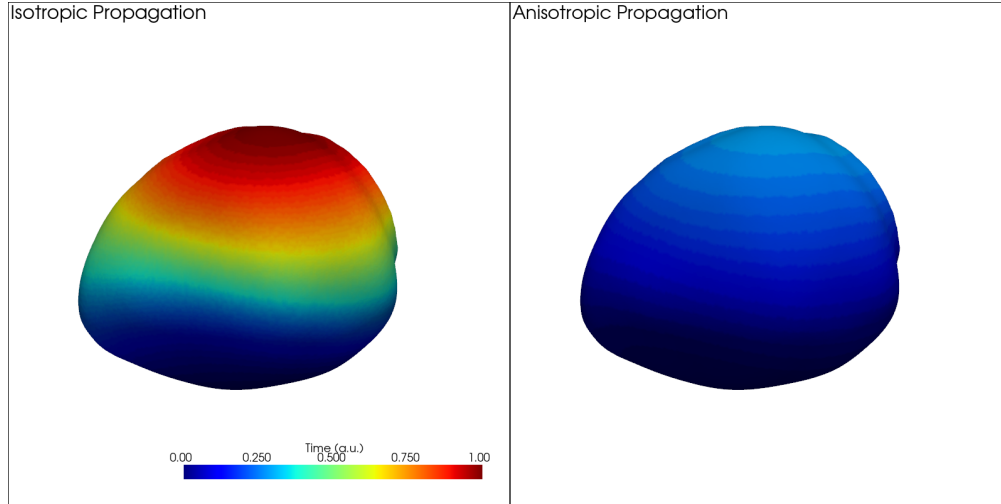


Figure 6: Side-by-side comparison of the activation maps on the realistic left-ventricular mesh: Isotropic conduction vs. Anisotropic conduction guided by the R-RBM fiber field.

The comparison in Figure 6 highlights the critical role of fiber architecture in cardiac electrophysiology:

1. *Wavefront Morphology*: In the isotropic model, the signal propagates as uniform concentric spheres, completely ignoring the myocardial structure. Conversely, the anisotropic model produces an elongated, elliptical wavefront. The propagation correctly aligns with the local fiber direction f , validating the geometric effectiveness of the algorithm.
2. *Conduction Efficiency*: Anisotropic propagation is significantly faster, the electrical signal exploits the longitudinal "highways" of the fibers, overcoming the slow, uniform spread characterizing the isotropic case.

5 Conclusions

In this work, a particular Laplace-Dirichlet Rule-Based Method called Rossi RBM was successfully implemented and validated for the generation of the cardiac fiber field, following the framework proposed by Piersanti et al. ([1]). By exploiting the Finite Element Method within the Firedrake environment, the algorithm was tested incrementally, successfully transitioning from simple geometries to more detailed and realistic anatomical left ventricle meshes.

The results show that the model is able to reproduce the physiological transmural rotation of the fibers from endocardium to epicardium. Furthermore, the electrophysiological simulations provided some functional validation, clearly highlighting the impact of the fiber architecture: while the isotropic case exhibited slower and non-physiological spherical propagation, the anisotropic tensor captured the preferential, higher-velocity electrical conduction along the local axis of the cardiomyocytes.

In conclusion, this project demonstrates that despite necessary simplifications—such as a rigid geometry and a simplified monodomain formulation—rule-based methods offer a robust, computationally efficient alternative to complex experimental imaging techniques.

References

- [1] Roberto Piersanti et al. “Modeling cardiac muscle fibers in ventricular and atrial electrophysiology simulations”. In: *Computer Methods in Applied Mechanics and Engineering* 373 (2021).
- [2] Simone Rossi et al. “Thermodynamically consistent orthotropic activation model capturing ventricular systolic wall thickening in cardiac electromechanics”. In: *European Journal of Mechanics-A/Solids* 48 (2014).